

TrES-1b

R-1819

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_220906 · created 2026-08-03T04:34:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459828.7154 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1765 +/- 0.0066
Transit depth (Rp/Rs)^2	3.11 +/- 0.23 [%]
Orbital Inclination (inc)	89.95 +/- 0.47
Airmass coefficient 1 (a1)	1.284 +/- 0.016
Airmass coefficient 2 (a2)	-0.1925 +/- 0.0093
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	None
Optimal Aperture	4.45
Optimal Annulus	8.63
Transit Duration (day)	0.1078 +/- 0.00074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1765 ± 0.0066 vs 0.1358 (deviation 5.9σ)
Duration fit vs published	0.1078 d vs 0.1042 d (deviation 4.69σ)
Mid-time O–C	-6.9 min
Depth SNR	25.6
Residual scatter	1.29 %

Light curve

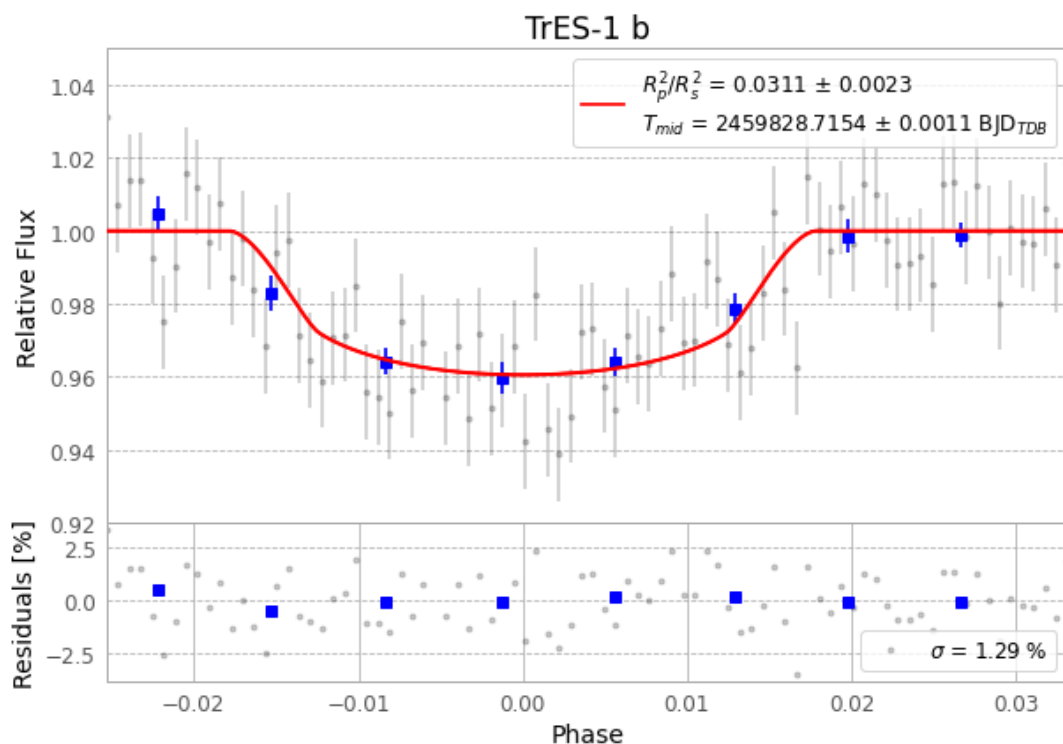


Figure 1 — FinalLightCurve_TrES-1 b_2022-09-05.png (SHA-256 7a58addde3ff8586...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 211], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 38d68cc9952e3cb3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), TRES-1220906031513.FITS → TRES-1220906073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-220906.log (7712ae1ec165...), FinalLightCurve_TrES-1 b_2022-09-05.png (7a58addde3ff...), FinalLightCurve_TrES-1 b_2022-09-05.csv (22a3abeec909...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:34Z.